

🚩 Flag 1 – Install Flutter SDK

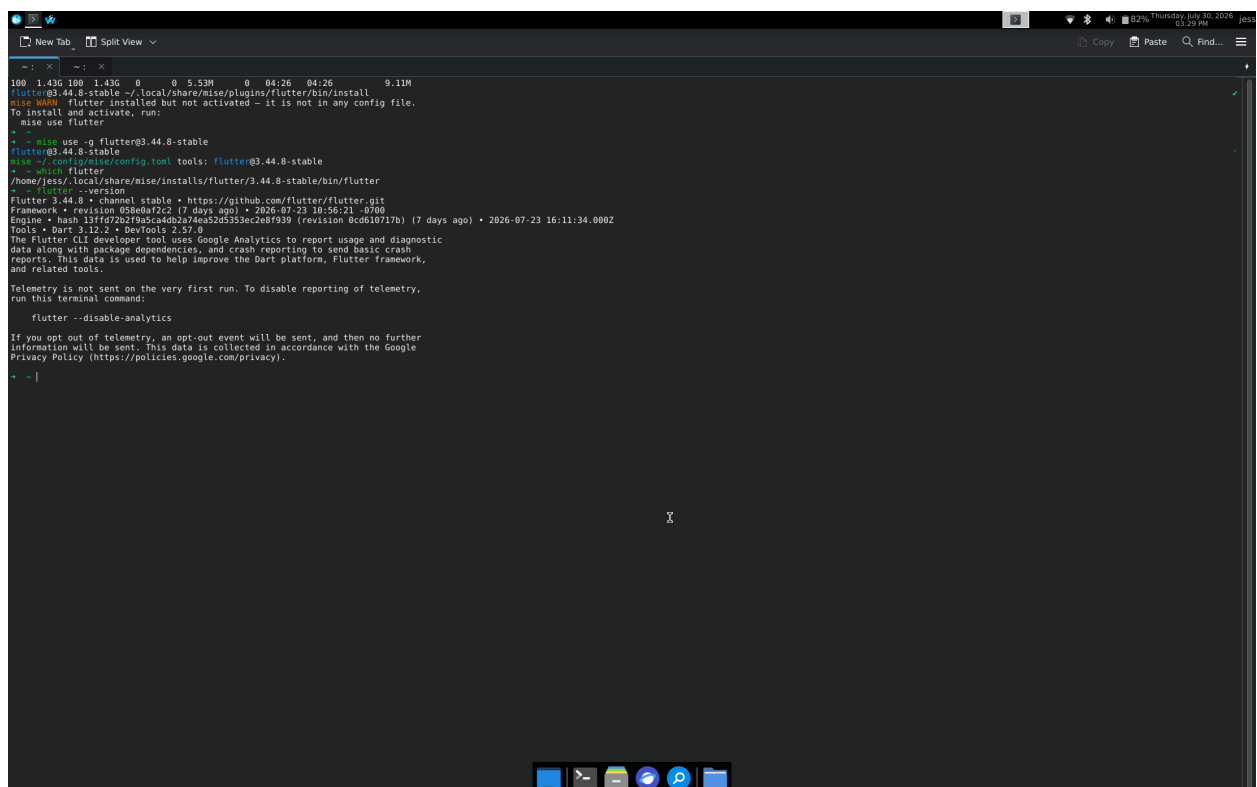
Objective: Install the Flutter SDK.

Verify:

```
flutter --version
```

Screenshot:

- The results of the command



```
100 1.43G 100 1.43G 0 0 5.53M 0 04:26 04:26 9.11M
Flutter 3.44.8-stable is already installed.
To install and activate, run:
  mise use flutter
+
+ - mise use -g flutter3.44.8-stable
Flutter 3.44.8-stable
+ use ~/config/mise/config.toml tools: flutter3.44.8-stable
+ - which flutter
/home/jes/.local/share/mise/installs/flutter/3.44.8-stable/bin/flutter
+ flutter --version
Flutter 3.44.8 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 03e0bf22 (7 days ago) • 2026-07-23 10:56:21 -0500
Engine • hash 13ff72b2f9a3c4d2a74ea52d5353ec2e8f939 (revision 0cd610717b) (7 days ago) • 2026-07-23 16:11:34.000Z
Tools • Dart 3.12.2 • DevTools 2.37.0
The Flutter CLI developer tool uses Google Analytics to report usage and diagnostic
data along with package dependencies, and crash reporting to send basic crash
reports. This data is used to help improve the Dart platform, Flutter framework,
and related tools.

Telemetry is not sent on the very first run. To disable reporting of telemetry,
run this terminal command:

  flutter --disable-analytics

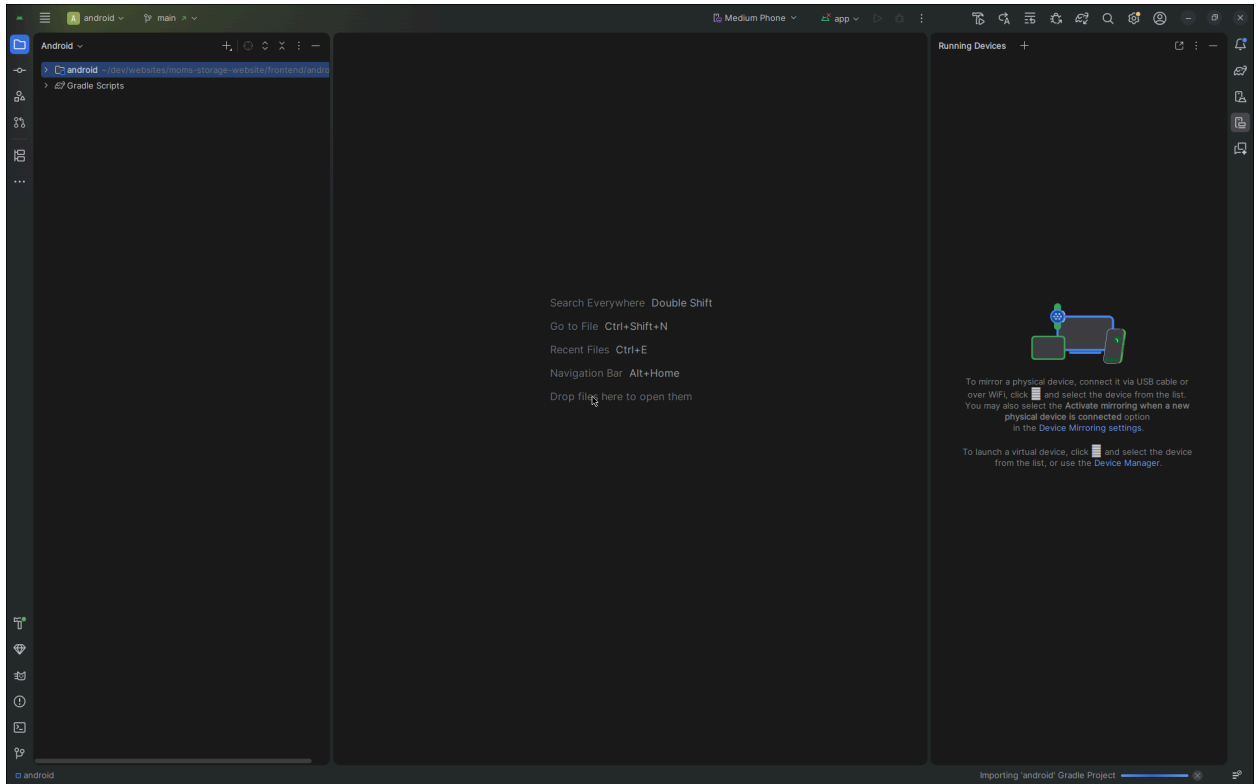
If you opt out of telemetry, an opt-out event will be sent, and then no further
information will be sent. This data is collected in accordance with the Google
Privacy Policy (https://policies.google.com/privacy).
```

🚩 Flag 2 – Install Android Studio

Objective: Install Android Studio.

Screenshot:

- Android Studio installed
- SDK Manager opened (or Welcome screen)



🚩 Flag 3 – Install VS Code & Flutter Extension

Objective: Install VS Code and the Flutter (which also installs Dart) extension.

Screenshot:

- Flutter extension installed
- Dart extension installed (or shown as installed)

The screenshot shows the Visual Studio Code interface with the 'Dart' extension page. The extension is by 'Dart Code' (dartcode.org) and has 15,452,702 downloads and a 4.8-star rating. It is described as 'Dart language support and debugger for Visual Studio Code.' The page includes sections for 'Introduction', 'Installation', 'Features', and 'Resources'. The 'Features' list includes: Edit and Debug Flutter mobile apps, Edit and Debug Dart command line apps, Automatic hot reloads for Flutter, Refactorings and Code fixes, Quickly switch between devices for Flutter, Flutter Doctor command, Flutter Get Packages command, Flutter Upgrade Packages command, Automatically gets packages when pubspec.yaml is saved, Automatically finds SDKs from PATH, Notification of new stable Dart SDK releases, Sort Members command, Support for debugging 'just my code' or SDK/libraries too, Prompts to get packages when out of date, Syntax highlighting, Code completion, Snippets, Realtime errors/warnings/TODOS, Documentation in hovers/tooltips, Go to Definition, Find References, Rename refactoring, Format document, Support for format-on-save, Support for format-on-type, and Workspace symbol search. The right sidebar shows 'Installation' details (Identifier: dart-code.dart-code, Version: 0.124.0, Last Updated: 15 minutes ago, Size: 14.8MB), 'Marketplace' info (Published: 10 years ago, Last Released: 2 weeks ago), 'Categories' (Programming Languages, Snippets, Linters, Formatters, Debuggers, Testing), and 'Resources' (Repository, Issues, License, Dart Code, Marketplace).

The screenshot shows the Visual Studio Code interface with the 'Flutter' extension page. The extension is by 'Dart Code' (dartcode.org) and has 34,380,308 downloads and a 4.9-star rating. It is described as 'Flutter support and debugger for Visual Studio Code.' The page includes sections for 'Introduction', 'Installation', 'Documentation', and 'Reporting Issues'. The 'Introduction' section states that this VS Code extension adds support for effectively editing, refactoring, running, and reloading Flutter mobile apps, and it depends on the Dart extension. The 'Installation' section says to install from the Visual Studio Code Marketplace or search within VS Code. The 'Documentation' section points to Flutter documentation. The 'Reporting Issues' section states that issues for both Dart and Flutter extensions should be reported in the Dart-Code issue tracker. The right sidebar shows 'Installation' details (Identifier: dart-code.flutter, Version: 0.124.0, Last Updated: 12 minutes ago, Size: 26.67KB), 'Marketplace' info (Published: 8 years ago, Last Released: 4 weeks ago), 'Categories' (Programming Languages, Snippets, Linters, Formatters, Debuggers), and 'Resources' (Repository, Issues, License, Dart Code, Marketplace).

🚩 Flag 4 – Run Flutter Doctor

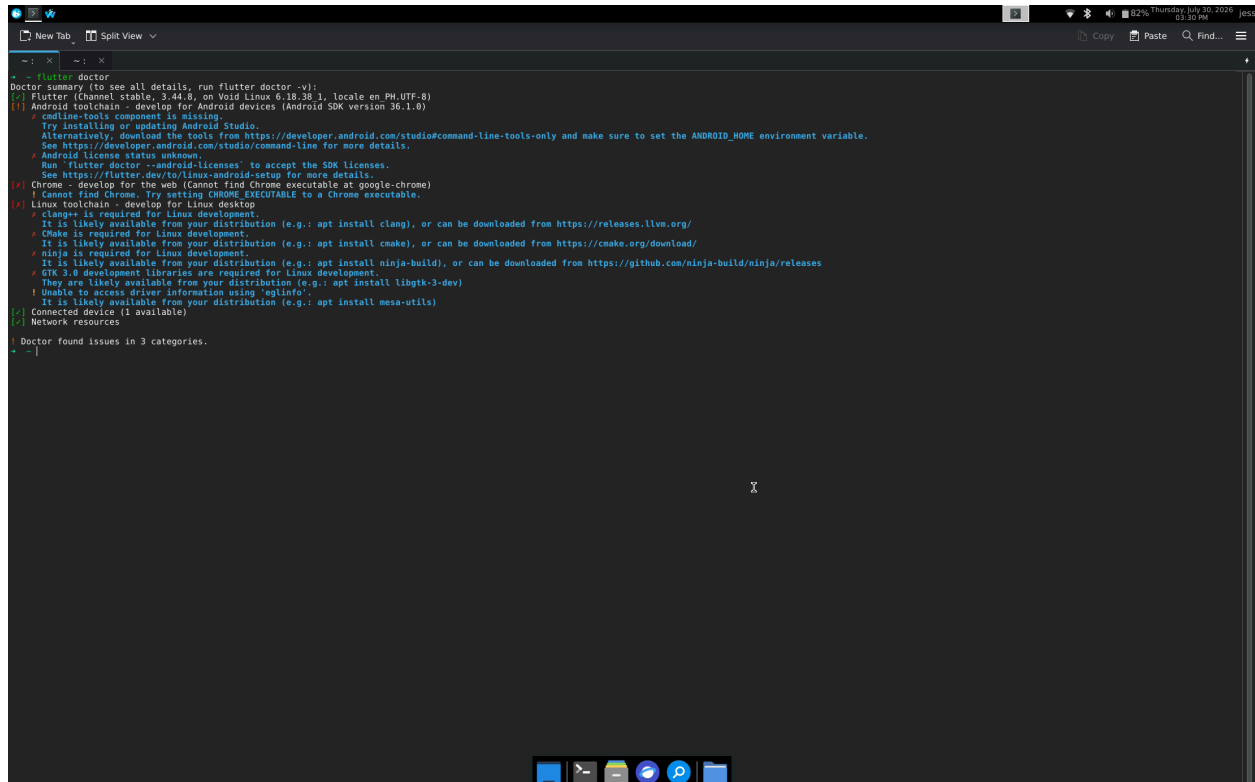
Objective: Check whether everything is configured correctly.

Run:

```
flutter doctor
```

Screenshot:

- flutter doctor results



```
flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 2.44.0, on Void Linux 6.18.35 1, locale en PM-UTF-8)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    ✗ cmdline-tools component is missing
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio/command-line and make sure to set the ANDROID_HOME environment variable.
      See https://developer.android.com/studio/command-line for more details.
    ✗ Android license status unknown.
      Run 'flutter doctor --android-licenses' to accept the SDK licenses.
      See https://flutter.dev/to/linux-android-setup for more details.
[✓] Chrome - develop for the web (Cannot find Chrome executable at google-chrome)
    ! Cannot find Chrome. Try setting CHROME_EXECUTABLE to a Chrome executable.
[✓] Linux toolchain - develop for Linux desktop
    ✗ clang++ is required for Linux development.
      It is likely available from your distribution (e.g.: apt install clang), or can be downloaded from https://releases.lldv.org/
    ✗ cmake is required for Linux development.
      It is likely available from your distribution (e.g.: apt install cmake), or can be downloaded from https://cmake.org/download/
    ✗ ninja is required for Linux development.
      It is likely available from your distribution (e.g.: apt install ninja-build), or can be downloaded from https://github.com/ninja-build/ninja/releases
    ✗ GTK 3.0 development libraries are required for Linux development.
      They are likely available from your distribution (e.g.: apt install libgtk-3-dev)
    ! Unable to access driver information using 'eglinfo'.
      It is likely available from your distribution (e.g.: apt install mesa-utils)
[✓] Connected device (1 available)
[✓] Network resources

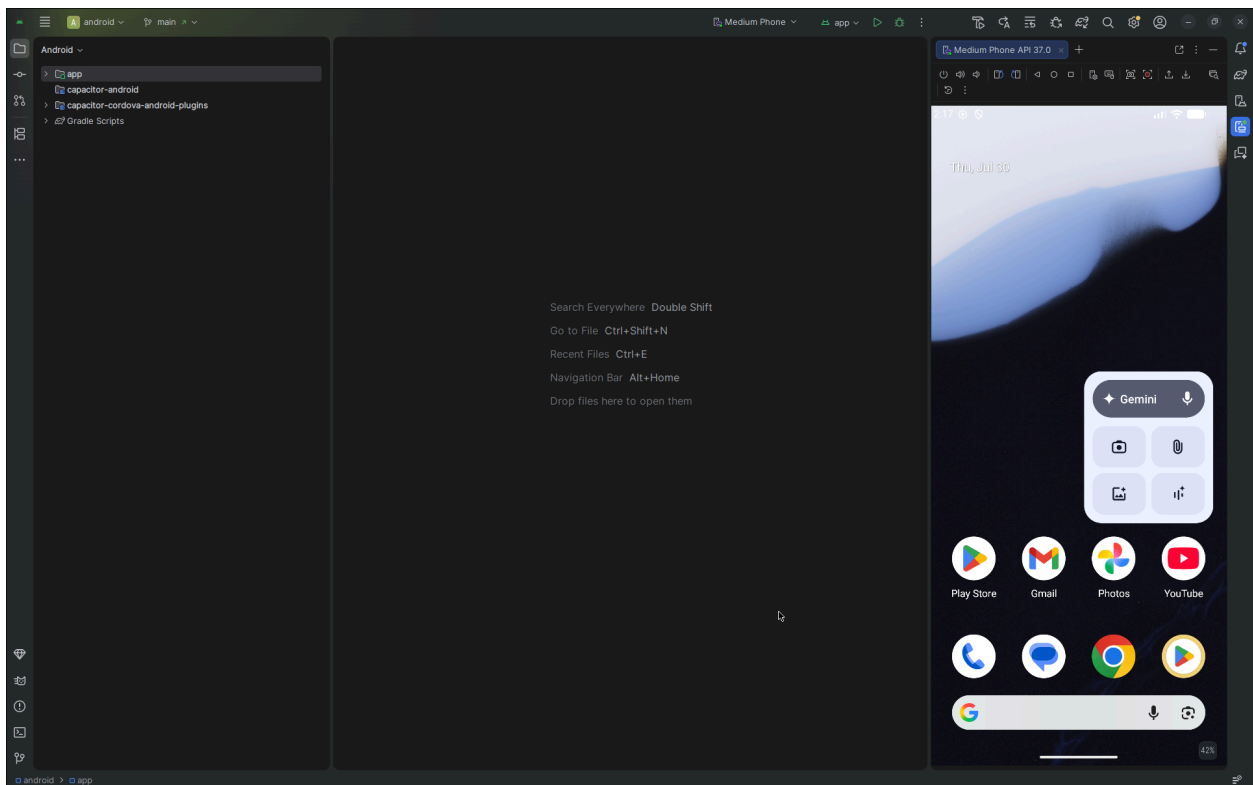
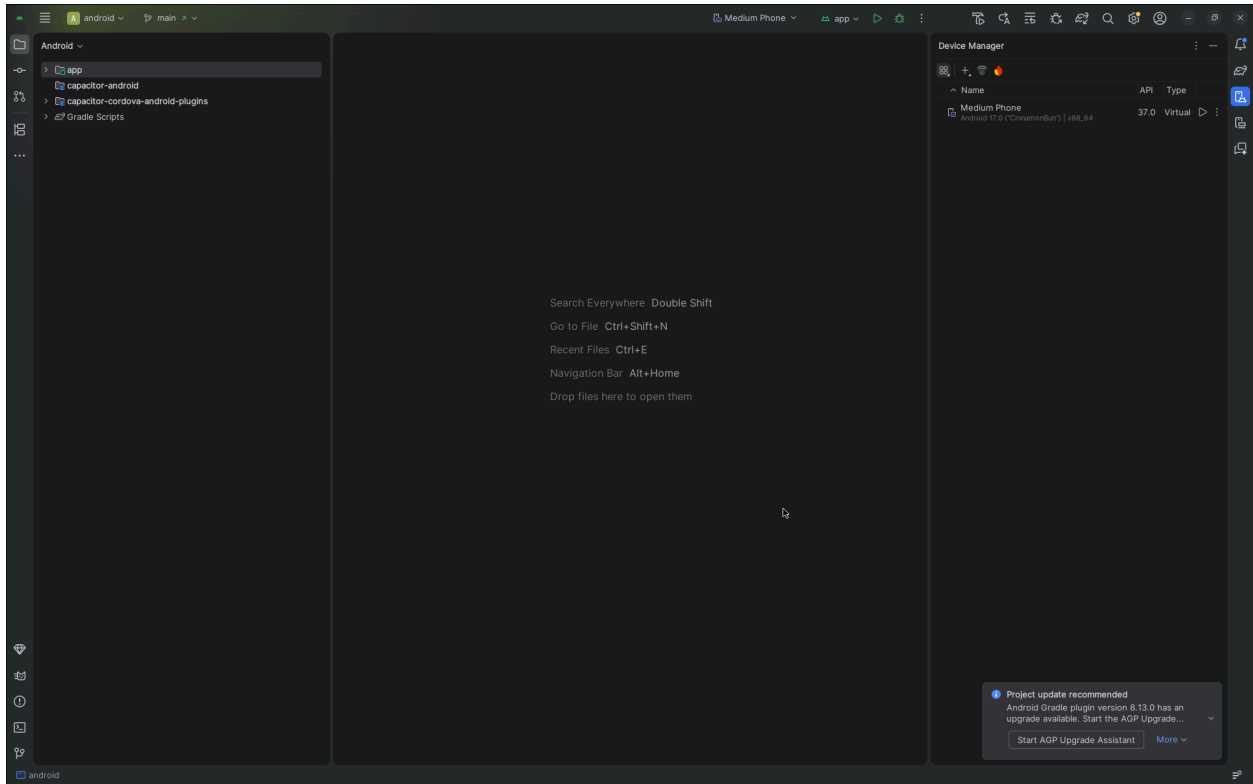
Doctor found issues in 3 categories.
```

Flag 5 – Build Your Virtual Device

Objective: Create your first Android Emulator (AVD).

Screenshot:

- Device Manager
- Your newly created virtual device
- Emulator successfully booted



🚩 Flag 6 – Forge Your Project

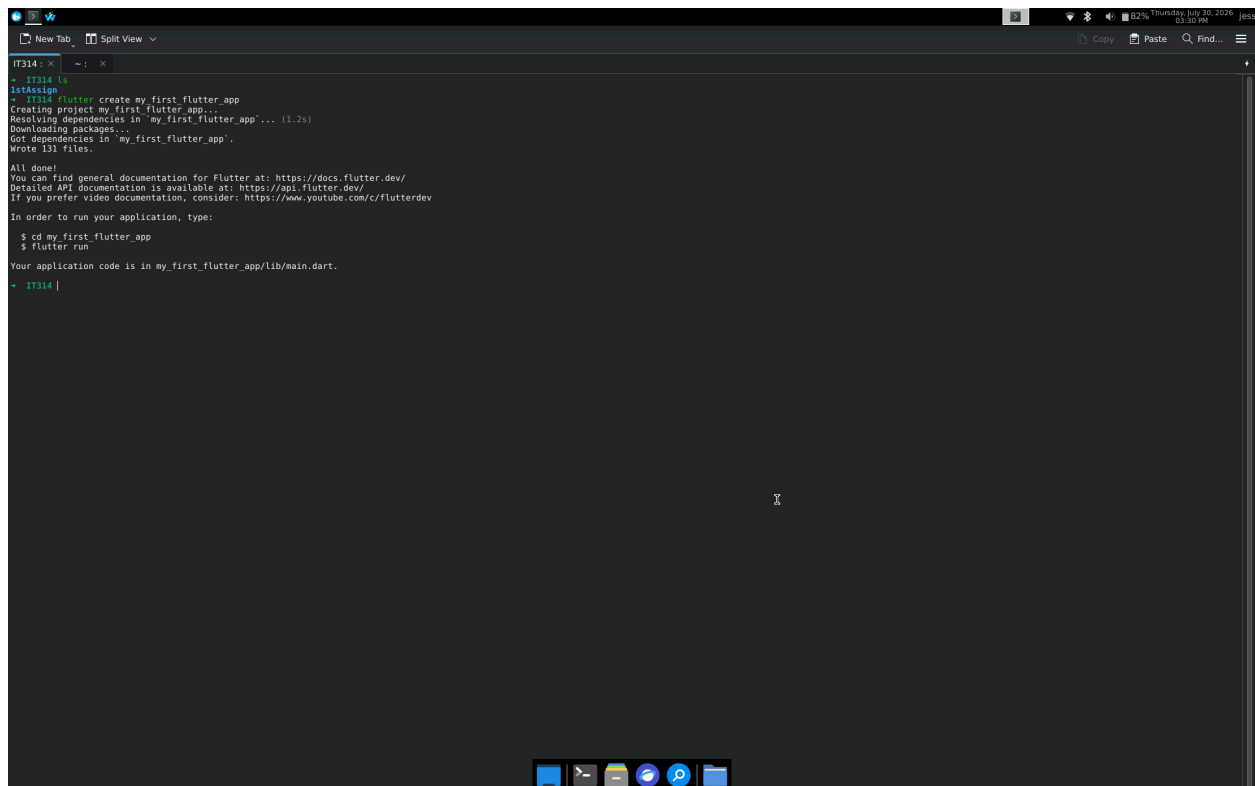
Objective: Create your first Flutter project.

Run:

```
flutter create my_first_flutter_app
```

Screenshot:

- Successful creation of project



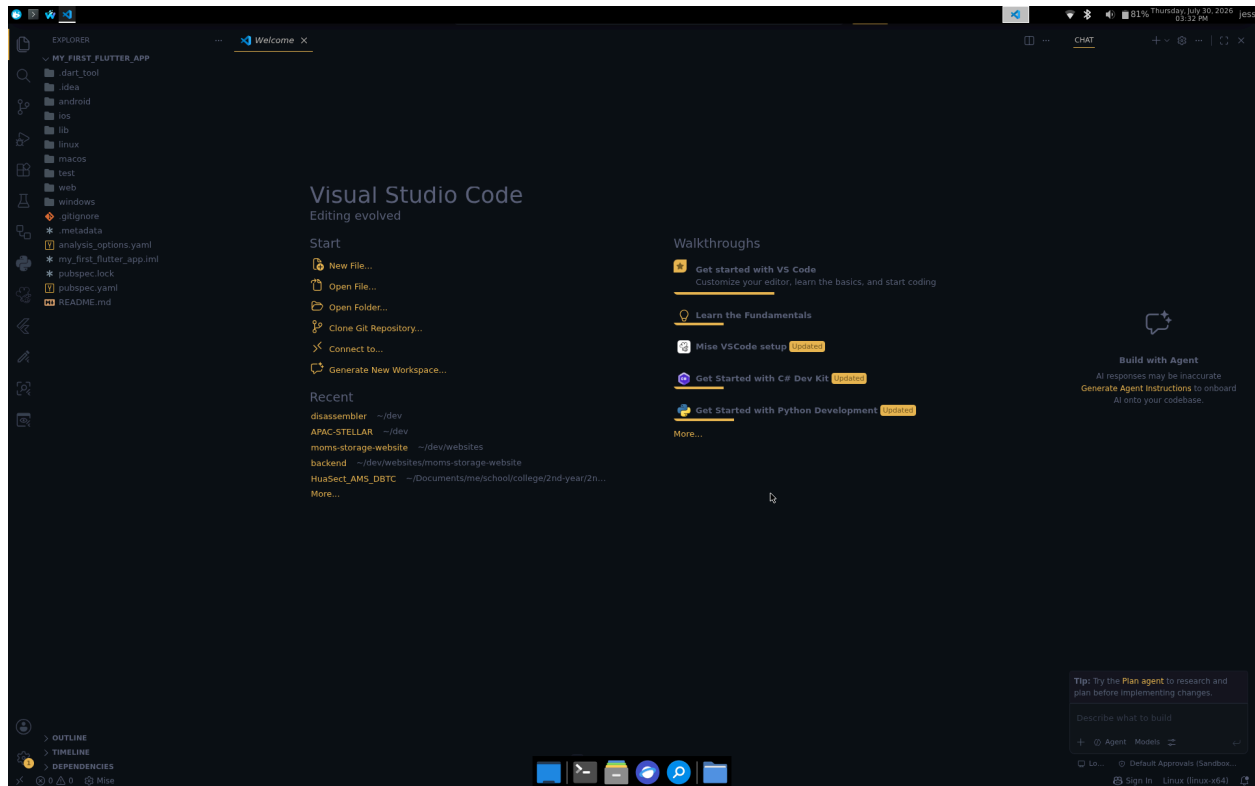
```
IT314: ~  
- IT314  
$ flutter create my_first_flutter_app  
Creating project my_first_flutter_app...  
Resolving dependencies in 'my_first_flutter_app'... (1.2s)  
Downloading packages...  
Got dependencies in 'my_first_flutter_app'.  
Wrote 131 files.  
  
All done!  
You can find general documentation for Flutter at: https://docs.flutter.dev/  
Detailed API documentation is available at: https://api.flutter.dev/  
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev  
  
In order to run your application, type:  
  
$ cd my_first_flutter_app  
$ flutter run  
  
Your application code is in my_first_flutter_app/lib/main.dart.  
- IT314 |
```

Flag 7 – Unlock the Workspace

Objective: Open the project in VS Code.

Screenshot:

- Your project opened in VS Code



🚩 Flag 8 – Launch

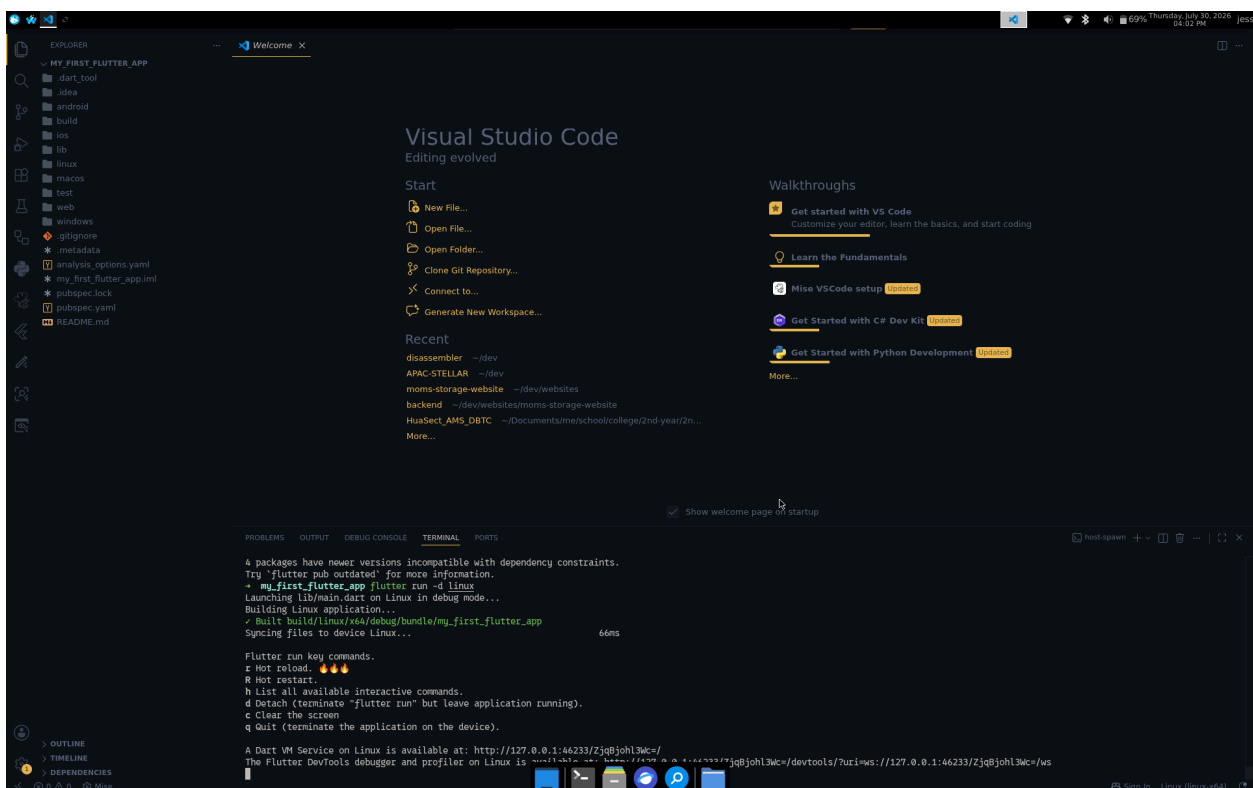
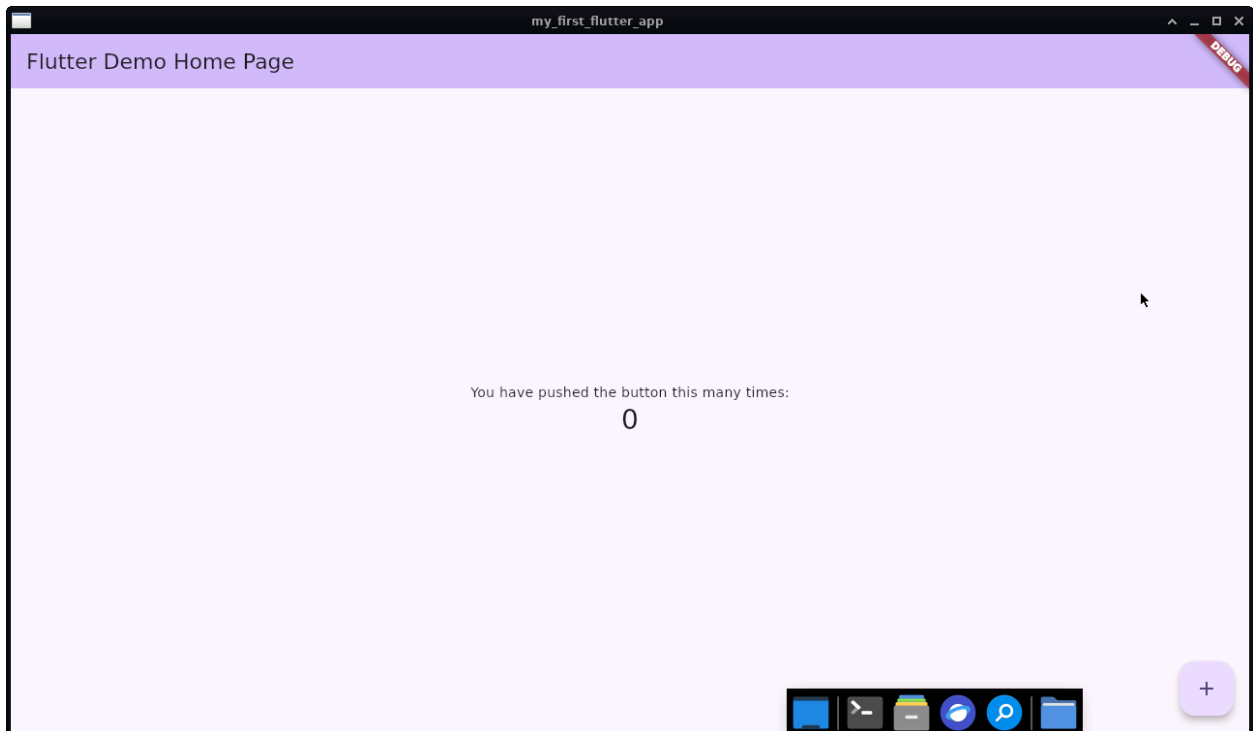
Objective: Run your Flutter application.

Run:

flutter run

Screenshot:

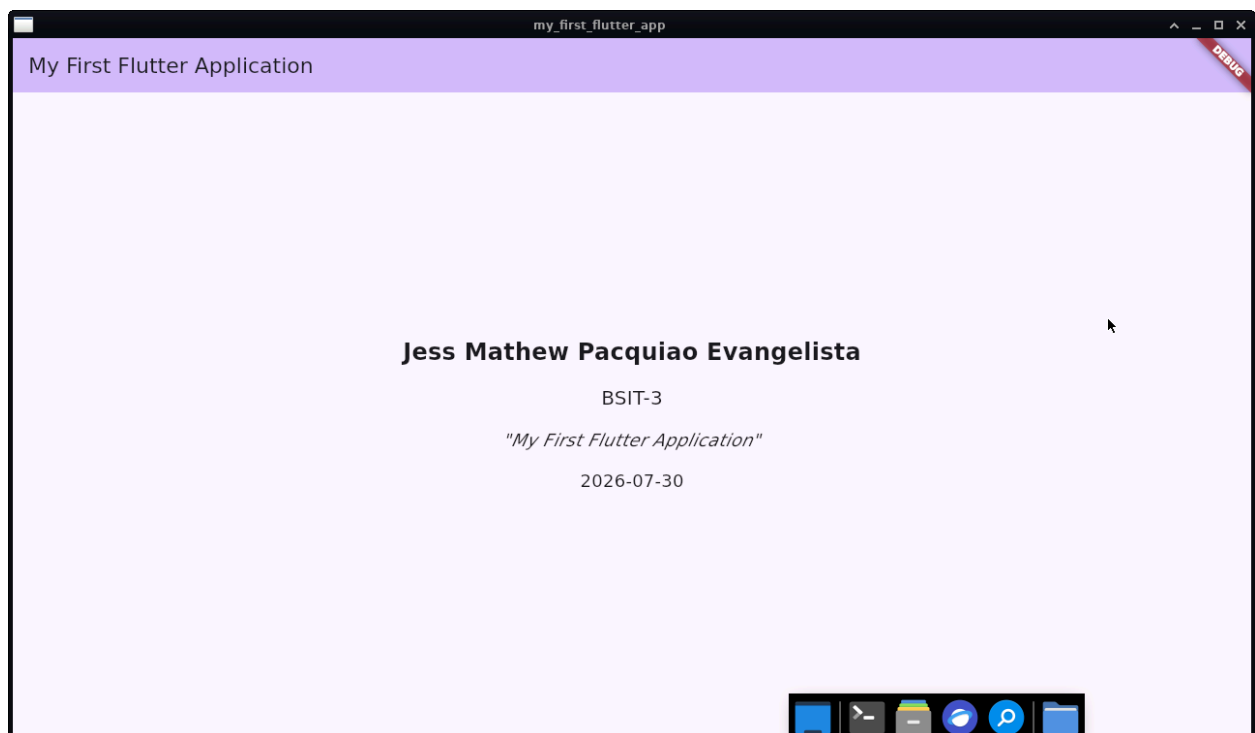
- Terminal
- Emulator running
- Default Flutter app



🚩 Flag 9 – Find the Entry Point

- Your updated content of the code
- The updated UI

```
lib > main.dart > MyHomePage > build
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  // This widget is the root of your application.
11  Widget build(BuildContext context) {
12    return MaterialApp(
13      title: 'My First Flutter Application',
14      theme: ThemeData(
15        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
16      ), // ThemeData
17      home: const MyHomePage(),
18    ); // MaterialApp
19  }
20 }
21
22 class MyHomePage extends StatelessWidget {
23   const MyHomePage({super.key});
24
25  // This widget is the home page of your application.
26  Widget build(BuildContext context) {
27    DateTime now = DateTime.now();
28    String date = "${now.year}-${now.month.toString().padLeft(2, '0')}-${now.day.toString().padLeft(2, '0')}";
29
30    return Scaffold(
31      appBar: AppBar(
32        backgroundColor: Theme.of(context).colorScheme.inversePrimary,
33        title: const Text('My First Flutter Application'),
34      ), // AppBar
35      body: Center(
36        child: Column(
37          mainAxisAlignment: MainAxisAlignment.center,
38          children: [
39            const Text(
40              'Jess Mathew Pacquiao Evangelista',
41              style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
42            ), // Text
43            const SizedBox(height: 16),
44            const Text(
45              'BSIT-3',
46              style: TextStyle(fontSize: 20),
47            ), // Text
48            const SizedBox(height: 16),
49            const Text(
50              'My First Flutter Application',
51              style: TextStyle(fontSize: 18, fontStyle: FontStyle.italic),
52            ), // Text
53            const SizedBox(height: 16),
54            Text(
55              date,
56              style: const TextStyle(fontSize: 18),
57            ), // Text
58          ], // Column
59        ),
60      ),
61    );
62  }
63 }
```



Flag 11 – Personalize It

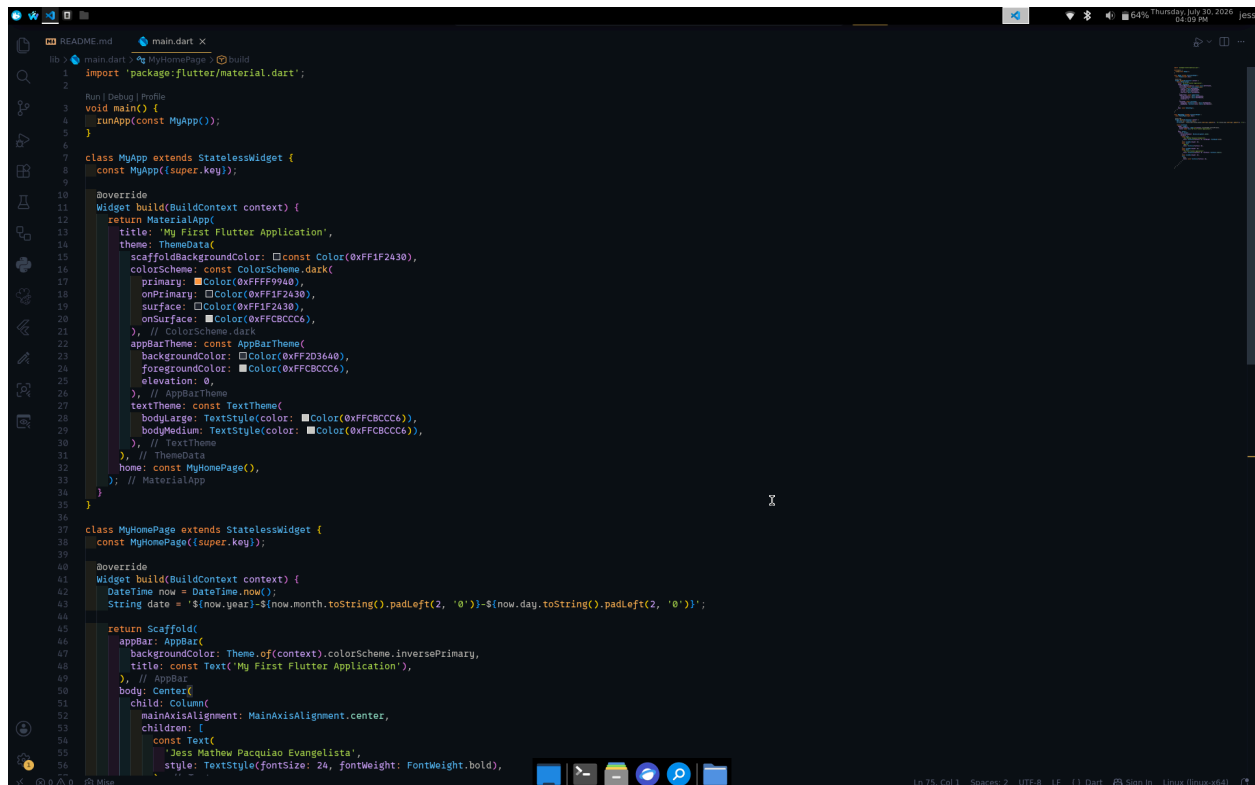
Research and implement at least **three** visual changes.

Examples:

- Change the background colour.
- Change the text colour.
- Centre the content.
- Increase font size.
- Add a border or rounded card.
- Change the font.
- Add an emoji or icon.

Screenshot:

- The code changed for personalizing
- The updated UI



```
lib > main.dart > MyHomePage > build
1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  // This widget is the root of your application.
11  Widget build(BuildContext context) {
12    return MaterialApp(
13      title: 'My First Flutter Application',
14      theme: ThemeData(
15        scaffoldBackgroundColor: const Color(0xFF1F2430),
16        colorScheme: const ColorScheme.dark(
17          primary: const Color(0xFFFF9940),
18          onPrimary: const Color(0xFF1F2430),
19          surface: const Color(0xFF1F2430),
20          onSurface: const Color(0xFFC8CC66),
21        ), // ColorScheme.dark
22        appBarTheme: const AppBarTheme(
23          backgroundColor: const Color(0xFF2D3640),
24          foregroundColor: const Color(0xFFC8CC66),
25          elevation: 0,
26        ), // AppBarTheme
27        textTheme: const TextTheme(
28          bodyLarge: TextStyle(color: const Color(0xFFC8CC66)),
29          bodyMedium: TextStyle(color: const Color(0xFFC8CC66)),
30        ), // TextTheme
31      ), // ThemeData
32      home: const MyHomePage(),
33    ); // MaterialApp
34  }
35 }
36
37 class MyHomePage extends StatelessWidget {
38   const MyHomePage({super.key});
39
40  // This widget represents the main content of the application.
41  Widget build(BuildContext context) {
42    DateTime now = DateTime.now();
43    String date = '${now.year}-${now.month.toString().padLeft(2, '0')}-${now.day.toString().padLeft(2, '0')}';
44
45    return Scaffold(
46      appBar: AppBar(
47        backgroundColor: Theme.of(context).colorScheme.inversePrimary,
48        title: const Text('My First Flutter Application'),
49      ), // AppBar
50      body: Center(
51        child: Column(
52          mainAxisAlignment: MainAxisAlignment.center,
53          children: [
54            const Text(
55              'Joss Matheu Pacquiao Evangelista',
56              style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
57            ),
58          ],
59        ),
60      ),
61    );
62  }
63 }
```



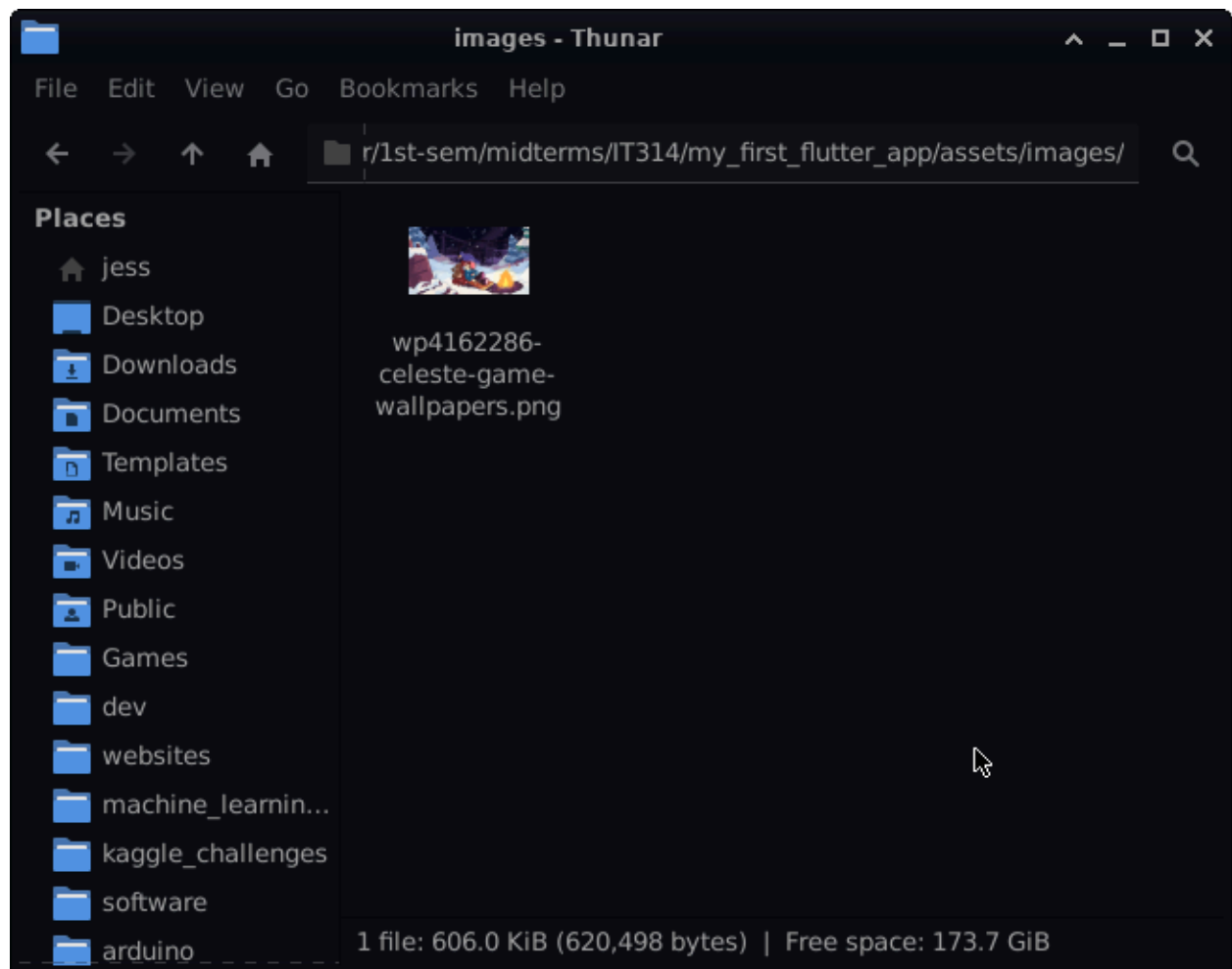
🚩 Flag 12 – Add an Image

Research how to:

- Create an `assets` folder
- Register assets
- Display an image

Screenshot:

- The steps it took to add an image
- The updated UI



```
EXPLORER
  MY_FIRST_FLUTTER_APP
    .dart_tool
    .idea
    android
    assets
    build
    ios
    lib
      main.dart
    linux
    macos
    test
    web
    windows
    .gitignore
    .metadata
    analysis_options.yaml
    my_first_flutter_app.iml
    pubspec.lock
    pubspec.yaml
    README.md

main.dart
  class MyApp extends StatelessWidget {
    Widget build(BuildContext context) {
      // ...
    }
  }

  class MyHomePage extends StatelessWidget {
    const MyHomePage({super.key});

    @override
    Widget build(BuildContext context) {
      DateTime now = DateTime.now();
      String date = '${now.year}-${now.month.toString().padLeft(2, '0')}-${now.day.toString().padLeft(2, '0')}';

      return Scaffold(
        appBar: AppBar(
          backgroundColor: Theme.of(context).colorScheme.inversePrimary,
          title: const Text('My First Flutter Application'),
        ), // AppBar
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Image.asset(
                'assets/images/wp4162286-celeste-game-wallpapers.png',
                height: 150,
                fit: BoxFit.contain,
              ), // Image.asset
              const SizedBox(height: 24),
              const Text(
                'Jess Mathew Pacquiao Evangelista',
                style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
              ), // Text
              const SizedBox(height: 16),
              const Text(
                'BSIT-3',
                style: TextStyle(fontSize: 20),
              ), // Text
              const SizedBox(height: 16),
              const Text(
                'My First Flutter Application',
                style: TextStyle(fontSize: 18, fontStyle: FontStyle.italic),
              ), // Text
              const SizedBox(height: 16),
              Text(
                date,
                style: const TextStyle(fontSize: 18),
              ), // Text
            ], // Column
          ), // Center
        ), // Scaffold
      );
    }
  }
}
```

